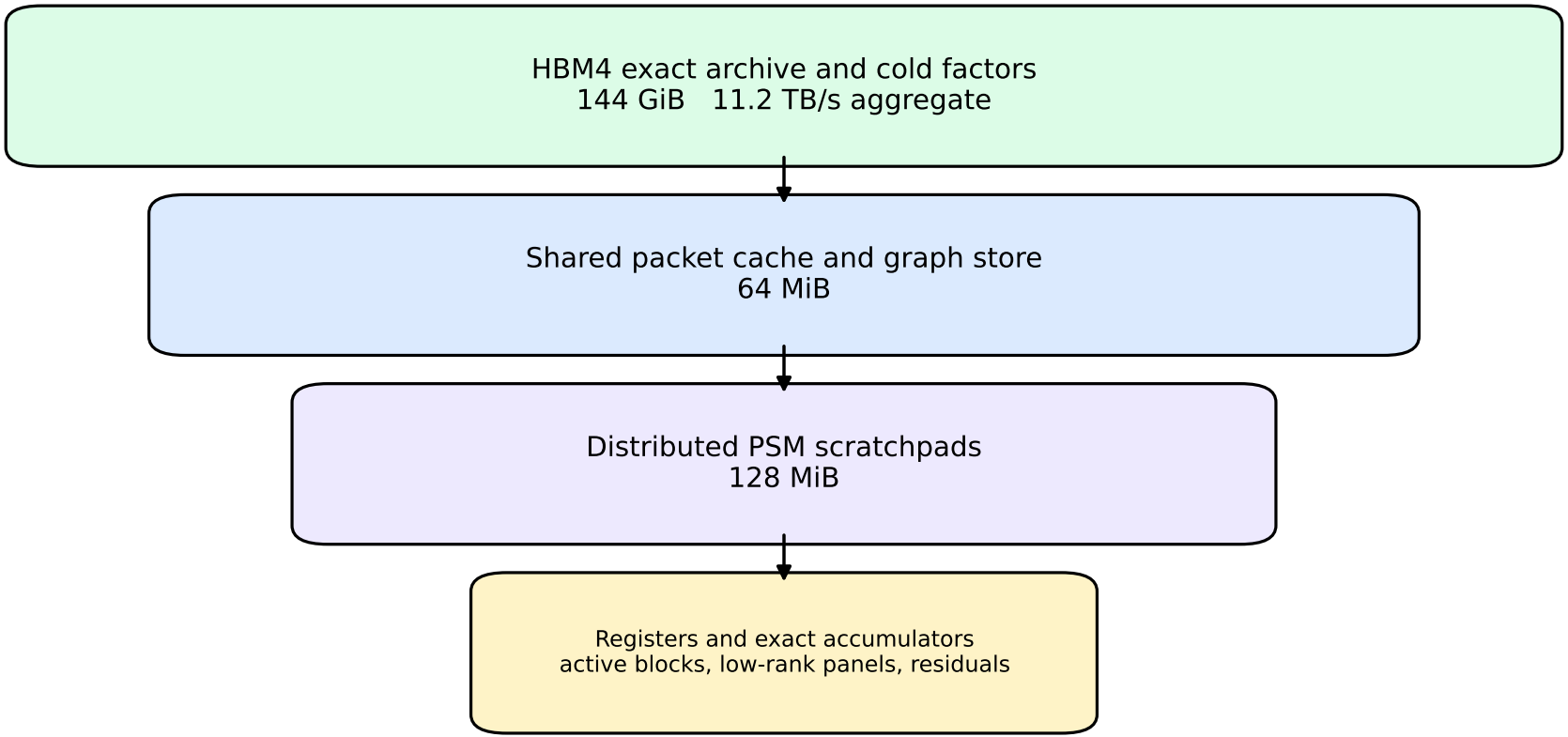
Factor-first packet memory hierarchy



No dense array is required. Explicit materialization is only one optional output contract.